

Representation Theory of $\mathrm{SL}_n(\mathbb{Z})$

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Abstract

We give a short summary of the irreducible representations of $\mathrm{SL}_n(\mathbb{Z})$ for $n \geq 3$ and prove the super-rigidity principle for $\mathrm{SL}_n(\mathbb{Z})$. The original proof is due to Bass, Milnor, and Serre's solution to the congruence subgroup problem for $\mathrm{SL}_n(\mathbb{Z})$ [BMS67] and uses the theory of analytic pro- p groups. We present the special case of that argument adapted to $\mathrm{SL}_n(\mathbb{Z})$.

1 Introduction

Let $\mathbf{k}$ be a field of characteristic zero and fix $n \geq 3$. Representations of $\mathrm{SL}_n(\mathbb{Z})$ are induced from two sources. Firstly, every rational representation of the algebraic group $\mathrm{SL}_n(\mathbf{k})$ restricts to a representation of $\mathrm{SL}_n(\mathbb{Z})$. Secondly, if $\pi : \mathrm{SL}_n(\mathbb{Z}) \rightarrow F$ is a surjective map into a finite group F and $\rho : F \rightarrow \mathrm{GL}_r(\mathbf{k})$ is a representation of F over $\mathbf{k}$, then $\rho \circ \pi : \mathrm{SL}_n(\mathbb{Z}) \rightarrow \mathrm{GL}_r(\mathbf{k})$ is a representation of $\mathrm{SL}_n(\mathbb{Z})$. By the universal property of profinite completion, representations of the second type factor through as continuous representations of the profinite completion $\widehat{\mathrm{SL}_n(\mathbb{Z})}$. And when $\mathbf{k}$ is algebraically closed, both types of representations are completely classified.

It turns out that these two sources of representations are all we need to classify the representation theory of $\mathrm{SL}_n(\mathbb{Z})$. The inclusion maps $\mathrm{SL}_n(\mathbb{Z}) \hookrightarrow \mathrm{SL}_n(\mathbf{k})$ and $\mathrm{SL}_n(\mathbb{Z}) \hookrightarrow \widehat{\mathrm{SL}_n(\mathbb{Z})}$ give an inclusion $\iota_0 : \mathrm{SL}_n(\mathbb{Z}) \hookrightarrow \mathrm{SL}_n(\mathbf{k}) \times \widehat{\mathrm{SL}_n(\mathbb{Z})}$. Define a rational representation of $\mathrm{SL}_n(\mathbf{k}) \times \widehat{\mathrm{SL}_n(\mathbb{Z})}$ to be a representation $\rho : \mathrm{SL}_n(\mathbf{k}) \times \widehat{\mathrm{SL}_n(\mathbb{Z})} \rightarrow \mathrm{GL}_r(\mathbf{k})$ such that $\rho|_{\mathrm{SL}_n(\mathbf{k})}$ is rational and $\rho|_{\widehat{\mathrm{SL}_n(\mathbb{Z})}}$ is continuous.

Putman [Putb] proves the following results which together completely classify the representation theory of $\mathrm{SL}_n(\mathbb{Z})$.

Theorem A. *Let $\rho : \mathrm{SL}_n(\mathbb{Z}) \rightarrow \mathrm{GL}_r(\mathbf{k})$ be a representation. Then ρ uniquely factors through a rational representation of $\mathrm{SL}_n(\mathbf{k}) \times \widehat{\mathrm{SL}_n(\mathbb{Z})}$, i.e. there is a unique rational representation $\hat{\rho} : \mathrm{SL}_n(\mathbf{k}) \times \widehat{\mathrm{SL}_n(\mathbb{Z})} \rightarrow \mathrm{GL}_r(\mathbf{k})$ such that $\rho = \hat{\rho} \circ \iota_0$. Moreover, any subspace $W \subseteq \mathbf{k}^n$ which is an $\mathrm{SL}_n(\mathbb{Z})$ -subrepresentation is also an $\mathrm{SL}_n(\mathbf{k}) \times \widehat{\mathrm{SL}_n(\mathbb{Z})}$ -subrepresentation.*

Theorem B. *Representations of $\mathrm{SL}_n(\mathbb{Z})$ decompose into direct sums of irreducibles.*

Theorem C. *Suppose $\mathbf{k}$ is algebraically-closed. Then the irreducible rational representations of $\mathrm{SL}_n(\mathbf{k}) \times \widehat{\mathrm{SL}_n(\mathbb{Z})}$ are precisely those of the form $V \otimes W$ where V is an irreducible rational representation of $\mathrm{SL}_n(\mathbf{k})$ and W is a continuous representation of $\widehat{\mathrm{SL}_n(\mathbb{Z})}$.*

In order to prove Theorem A, Putman assumes a general super-rigidity result of Margulis to derive:

Theorem D. *Let $\rho : \mathrm{SL}_n(\mathbb{Z}) \rightarrow \mathrm{GL}_r(\mathbf{k})$ be a representation. Then there exists a finite-index subgroup $\Gamma \leq \mathrm{SL}_n(\mathbb{Z})$ and a rational representation $\bar{\rho} : \mathrm{SL}_n(\mathbf{k}) \rightarrow \mathrm{GL}_r(\mathbf{k})$ such that $\rho|_{\Gamma} = \bar{\rho}|_{\Gamma}$.*

In this document we provide an alternate proof of Theorem D by instead using the theory of analytic pro- p groups, as described by Bass, Milnor, and Serre [BMS67]. We also demonstrate a different way to derive the existence half of Theorem A from Theorem D. Refer to [Putb] for self-contained proofs of Theorem B, Theorem C, and the uniqueness half of Theorem A.

2 Preliminaries

We assume familiarity with the theory of analytic pro- p groups for $p > 2$ prime, e.g. up to Chapter 9 of [DDSSM09]. We will also take for granted the representation theory of $\mathrm{SL}_n(\mathbf{k}')$ and $\mathfrak{sl}_n(\mathbf{k}')$ for fields $\mathbf{k}'$ of characteristic zero.

To investigate the representations of $\widehat{\mathrm{SL}_n(\mathbb{Z})}$, we make use of the Congruence Subgroup Property for $\mathrm{SL}_n(\mathbb{Z})$.

Theorem 1 (Congruence Subgroup Property (CSP)). *Let $\Gamma_n(\ell) := \ker(\mathrm{SL}_n(\mathbb{Z}) \rightarrow \mathrm{SL}_n(\mathbb{Z}/\ell\mathbb{Z}))$. Given a finite index subgroup K of $\mathrm{SL}_n(\mathbb{Z})$, there exists $m \in \mathbb{N}$ such that $\Gamma_n(m) \leq K$.*

Proof. We refer the reader to Putman's proof. [Puta] □

This implies that

$$\widehat{\mathrm{SL}_n(\mathbb{Z})} \cong \varprojlim_m \mathrm{SL}_n(\mathbb{Z}/m\mathbb{Z}) \cong \mathrm{SL}_n(\hat{\mathbb{Z}}).$$

By the Chinese Remainder Theorem, we also obtain that

$$\widehat{\mathrm{SL}_n(\mathbb{Z})} \cong \prod_p \mathrm{SL}_n(\mathbb{Z}_p)$$

as topological groups, where the product is taken over primes p .

We will also need some facts about $\mathrm{SL}_n(\mathbb{Z}_p)$ and $\mathrm{GL}_r(\mathbb{Z}_p)$ for primes p . These are p -adic analytic groups, meaning that any of their open subgroups contain

a further open uniform pro- p subgroup; in fact, these uniform pro- p subgroups can be taken to be one of the principal congruence subgroups.

Under the equivalence of categories between local p -adic analytic groups and $\mathbb{Q}_p$ -Lie algebras, $\mathrm{SL}_n(\mathbb{Z}_p)$ corresponds to $\mathfrak{sl}_n(\mathbb{Q}_p)$ and $\mathrm{GL}_r(\mathbb{Z}_p)$ corresponds to $\mathfrak{gl}_r(\mathbb{Q}_p)$.

3 Super-rigidity proof

In this section, we will prove Theorem D. We will need the following lemma.

Lemma 1. *Let R an integral domain of characteristic 0 which is finitely generated as a $\mathbb{Z}$ -module. Then there are infinitely many primes $p \in \mathbb{Z}$ such there exists a ring embedding $R \hookrightarrow \mathbb{Z}_p$.*

Proof. This follows from Cassels' embedding theorem [Cas76]. Let $r_1, \dots, r_m$ be a set of nonzero generators for R and let F be the field of fractions of R . By Cassels' embedding theorem, there are infinitely many primes p for which we have an embedding $\alpha : F \rightarrow \mathbb{Q}_p$ such that $|\alpha(r_i)| \leq 1$ for all i . Then clearly $\alpha|_R$ is a ring embedding that maps into $\mathbb{Z}_p$. $\square$

Proof of Theorem D. $\mathrm{SL}_n(\mathbb{Z})$ is finitely-generated, so let $A_1, \dots, A_m$ be generators. ρ is determined by its values on the A_i , so if R is the subring of $\mathbf{k}$ generated by $\mathbb{Z}$ as well as the matrix entries of $\rho(A_i)$ for all i , then ρ maps into $\mathrm{GL}_r(R)$. By Lemma 1, pick a prime $p > 2$ such that R embeds in $\mathbb{Z}_p$, so we can view ρ as a map $\rho : \mathrm{SL}_n(\mathbb{Z}) \rightarrow \mathrm{GL}_r(\mathbb{Z}_p)$.

Since the codomain is a profinite group, this map factors through a continuous homomorphism

$$\rho' : \widehat{\mathrm{SL}_n(\mathbb{Z})} \rightarrow \mathrm{GL}_r(\mathbb{Z}_p)$$

such that the below diagram commutes.

$$\begin{array}{ccc} \mathrm{SL}_n(\mathbb{Z}) & \xrightarrow{\rho} & \mathrm{GL}_r(\mathbb{Z}_p) \\ \downarrow \iota & \nearrow \rho' & \\ \widehat{\mathrm{SL}_n(\mathbb{Z})} & & \end{array}$$

Recall that

$$\widehat{\mathrm{SL}_n(\mathbb{Z})} \cong \prod_q \mathrm{SL}_n(\mathbb{Z}_q).$$

Let Γ_1 be the kernel of the surjection $\mathrm{GL}_r(\mathbb{Z}_p) \rightarrow \mathrm{GL}_r(\mathbb{Z}/p\mathbb{Z})$, which is an open uniform pro- p subgroup of $\mathrm{GL}_r(\mathbb{Z}_p)$. Let $\Gamma' = (\rho')^{-1}(\Gamma_1) \subseteq \prod_q \mathrm{SL}_n(\mathbb{Z}_q)$, which is an open and finite-index subgroup. By the product topology, there is an open subset Γ'' of $\prod_q \mathrm{SL}_n(\mathbb{Z}_q)$ such that

$$\Gamma'' := \prod_q U_q \subseteq \Gamma'$$

where U_q is an open compact subgroup of $\mathrm{SL}_n(\mathbb{Z}_q)$ and $U_q = \mathrm{SL}_n(\mathbb{Z}_q)$ for all but finitely many q . By using the congruence subgroup property for $\mathrm{SL}_n(\mathbb{Z}_p)$, we can assume that $U_p := \ker(\mathrm{SL}_n(\mathbb{Z}_p) \rightarrow \mathrm{SL}_n(\mathbb{Z}_p/p^k\mathbb{Z}_p))$ for some large k . Note that U_p is a uniform pro- p group with this assumption.

Lemma 2. $\rho'|_{U_q} : U_q \rightarrow \Gamma_1$ has trivial image for every $q \neq p$.

Proof. We first show the image of ρ' is finite. Since U_q is an open subgroup of $\mathrm{SL}_n(\mathbb{Z}_q)$, it is a q -adic analytic group. Let $K \leq U_q$ be some uniform pro- q subgroup. Then $\rho'(K) \subseteq \Gamma_1$ is a pro- q group, being the continuous image of K ; and is a closed (compact) subgroup of the pro- p group Γ_1 . Hence $\rho'(K)$ is also pro- p . So $\rho'(K) = \{1\}$. Now, since U_q is compact and K is open in U_q , we can cover U_q with finitely many translations of K . Thus, the image $\rho'(U_q)$ is a finite subgroup of Γ_1 .

We are now done, because Γ_1 is uniform and thus torsion-free. Hence, it has no non-trivial finite subgroups. $\square$

Let $\pi_p : \Gamma'' \rightarrow U_p$ be the projection map and $\tilde{\rho} := \rho'|_{U_p}$. Then $\rho' = \tilde{\rho} \circ \pi_p$ by a pointwise convergence argument, by Lemma 2 and because ρ' is continuous.

Let $\Gamma = \iota^{-1}(\Gamma'')$ which is a finite-index subgroup of $\mathrm{SL}_n(\mathbb{Z})$. Then we have shown that $\rho|_{\Gamma} : \Gamma \rightarrow \Gamma_1$ extends to a map $\tilde{\rho} : U_p \rightarrow \Gamma_1$, i.e. that the following diagram commutes.

$$\begin{array}{ccc} \Gamma & \xrightarrow{\rho|_{\Gamma}} & \Gamma_1 \\ \downarrow \iota & & \uparrow \tilde{\rho} \\ \prod_q U_q & \xrightarrow{\pi_p} & U_p \end{array}$$

Recall that Γ_1 and U_p are both uniform pro- p groups. In the language of Lazard's theory [Laz65], $\tilde{\rho}$ represents a germ of $\mathcal{H}(\mathrm{SL}_n(\mathbb{Z}_p), \Gamma_1)$, the set of all analytic homomorphisms whose domain is an open subgroup of $\mathrm{SL}_n(\mathbb{Z}_p)$ and whose codomain is Γ_1 . Via the equivalence of categories between local p -adic analytic groups and finite dimensional Lie algebras over $\mathbb{Q}_p$, we obtain a corresponding Lie algebra representation over $\mathbb{Q}_p$:

$$(\tilde{\rho})^* : \mathfrak{sl}_n(\mathbb{Q}_p) \rightarrow \mathfrak{gl}_r(\mathbb{Q}_p).$$

Now $\mathbb{Q}_p$ has characteristic 0, so such representations are completely classified: they decompose into direct sums of irreducible representations, and irreducible representations are given by applying Schur functors to the standard representation. So our Lie algebra representation $(\tilde{\rho})^*$ can be written as

$$\bigoplus_{\lambda_i} \mathbb{S}_{\lambda_i}(\mathbb{Q}_p^n) \tag{1}$$

for some collection of Young tableaux λ_i .

However, the same expression in (1) defines an $\mathrm{SL}_n(\mathbb{Q}_p)$ -representation $\varphi : \mathrm{SL}_n(\mathbb{Q}_p) \rightarrow \mathrm{GL}(V)$, whose induced Lie algebra representation is also $(\tilde{\rho})^*$. So by [DDSSM09] Theorem 9.11, we have that $\tilde{\rho} : U_p \rightarrow \Gamma_1$ and φ agree on any

uniform open subgroup around $1 \in \mathrm{SL}_n(\mathbb{Q}_p)$ on which they are both defined; in particular, they agree on U_p . So φ is an extension of $\tilde{\rho}$, and φ is a rational map of algebraic groups where the polynomials are given by integer coefficients. Hence $\tilde{\rho}$ is also rational with integer coefficients.

Now, the inclusion and projection $\Gamma \xrightarrow{\iota} \Gamma'' \xrightarrow{\pi_p} U_p$ are also rational with integer coefficients, hence our original representation $\rho|_\Gamma : \Gamma \rightarrow \Gamma_1$ is rational with integer coefficients. But recall that the codomain of ρ can in fact be taken to be $\mathrm{GL}_r(\mathbf{k})$. And polynomials with integer coefficients can take inputs and outputs in $\mathbf{k}$. Hence, using the integer coefficient polynomials defining $\rho|_\Gamma$, we can extend to a rational map $\bar{\rho} : \mathrm{SL}_n(\mathbf{k}) \rightarrow \mathrm{GL}_r(\mathbf{k})$, where $\rho|_\Gamma = \bar{\rho}|_\Gamma$. $\square$

We have now proven Theorem D. Putman's [Putb] proof of Theorem A suffices to conclude here, but we present here an alternative argument for the existence half of Theorem A.

Lemma 3. *Let G be a group and $H \leq G$ be a finite-index subgroup. Let $\rho : G \rightarrow \mathrm{GL}(V)$ be a representation. Then ρ is a quotient of $\mathrm{Ind}_H^G \mathrm{Res}_H^G(\rho)$.*

Proof. Let $W = \mathrm{Res}_H^G \rho$. By adjunction,

$$\mathrm{Hom}_G(\mathrm{Ind}_H^G W, V) \cong \mathrm{Hom}_H(W, V) \neq 0.$$

In particular, there exists a surjective map $f \in \mathrm{Hom}_H(W, V)$. Left adjoints preserve epimorphisms, hence there is a corresponding surjective map $\mathrm{Ind}_H^G f \in \mathrm{Hom}_G(\mathrm{Ind}_H^G W, V)$. (Alternatively, one can take f to be the identity in W and calculate $\mathrm{Ind}_H^G f$ explicitly under Hom-tensor adjunction to see that it is surjective.) Hence ρ is a quotient of $\mathrm{Ind}_H^G \mathrm{Res}_H^G(\rho)$. $\square$

Theorem 2. *Keep the same notations as in Theorem D. Then ρ factors through a rational representation $\hat{\rho} : \widehat{\mathrm{SL}_n(\mathbb{Z})} \times \mathrm{SL}_n(\mathbf{k}) \rightarrow \mathrm{GL}_r(\mathbf{k})$.*

Proof. Let $\psi = \mathrm{Ind}_\Gamma^{\mathrm{SL}_n(\mathbb{Z})} \rho|_\Gamma$. By Lemma 3, $\rho(g) = \pi \circ \psi(g)$ for all $g \in \mathrm{SL}_n(\mathbb{Z})$ for some surjective representation morphism π . It thus suffices to prove the result for ψ , i.e. that ψ factors through a rational representation $\hat{\psi}$ of $\widehat{\mathrm{SL}_n(\mathbb{Z})} \times \mathrm{SL}_n(\mathbf{k})$, and then setting $\hat{\rho}(g) = \pi \circ \hat{\psi}(g)$.

Let $t_1, \dots, t_d$ be a system of coset representatives of $\mathrm{SL}_n(\mathbb{Z})/\Gamma$. For $g \in \mathrm{SL}_n(\mathbb{Z})$, let $j_g : \{1, \dots, d\} \rightarrow \{1, \dots, d\}$ be the permutation such that for all i , $j_g(i)$ is the unique number with $t_{j_g(i)}^{-1} g t_i \in \Gamma$. Then $\psi(g)$ is the block-permutation matrix whose ij block is zero if $j \neq j_g(i)$ and equal to $\rho(t_{j_g(i)}^{-1} g t_i)$ otherwise.

Let $C(\Gamma) = \bigcap_{g \in \mathrm{SL}_n(\mathbb{Z})} g \Gamma g^{-1}$ be the normal core of Γ . Notice that j_g depends only on the image of g in $\mathrm{SL}_n(\mathbb{Z})/C(\Gamma)$, because if $h \in C(\Gamma)$ then $t_{j_g(i)}^{-1} g h t_i = t_{j_g(i)}^{-1} g t_i t_i^{-1} h t_i \in \Gamma$.

We will now define a representation $\hat{\psi}$ of $\widehat{\mathrm{SL}_n(\mathbb{Z})}/C(\Gamma) \times \mathrm{SL}_n(\mathbf{k})$, which automatically factors through a continuous representation of $\widehat{\mathrm{SL}_n(\mathbb{Z})} \times \mathrm{SL}_n(\mathbf{k})$.

Given $(gC(\Gamma), A) \in \mathrm{SL}_n(\mathbb{Z})/C(\Gamma) \times \mathrm{SL}_n(\mathbf{k})$, define $\hat{\psi}(gC(\Gamma), A)$ to be the permutation matrix whose nonzero blocks are $\bar{\rho}(t_{j_g(i)}^{-1}At_i)$, where $\bar{\rho}$ is as in Theorem

D. We show that $\hat{\psi}$ is a group homomorphism.

We first show $g \mapsto j_g$ is a homomorphism. This is true because

$$t_{j_g(j_h(i))}^{-1}ght_i = t_{j_g(j_h(i))}^{-1}gt_{j_h(i)}t_{j_h(i)}^{-1}ht_i \in \Gamma$$

and hence $j_{gh}(i) = j_g(j_h(i))$.

Let E_{ij} be the block matrix whose ij -block is the $n \times n$ identity matrix and whose other blocks are 0. Let $\otimes$ denote the matrix Kronecker product. Compute

$$\begin{aligned} & \hat{\psi}(gC(\Gamma), A)\hat{\psi}(hC(\Gamma), B) \\ &= \left(\sum_i E_{j_g(i)i} \otimes \bar{\rho}(t_{j_g(i)}^{-1}At_i) \right) \left(\sum_l E_{j_h(l)l} \otimes \bar{\rho}(t_{j_h(l)}^{-1}Bt_l) \right) \\ &= \sum_{i,l} (E_{j_g(i)i} \otimes \bar{\rho}(t_{j_g(i)}^{-1}At_i)) (E_{j_h(l)l} \otimes \bar{\rho}(t_{j_h(l)}^{-1}Bt_l)) \\ &= \sum_i E_{j_g(i)j_h^{-1}(i)} \otimes \bar{\rho}(t_{j_g(i)}^{-1}ABt_{j_h^{-1}(i)}) \\ &= \hat{\psi}(ghC(\Gamma), AB). \end{aligned}$$

Therefore $\hat{\psi}$ is a homomorphism. Also

$$\hat{\psi}|_{\mathrm{SL}_n(\mathbf{k})}(C(\Gamma), A) = \sum_i E_{ii} \otimes \bar{\rho}(t_i^{-1}At_i)$$

which is rational because $\bar{\rho}$ is rational. And $\hat{\psi}$ clearly extends ψ . Therefore we are done. $\square$

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